

Multi-Asset Risk Parity with Regime-Conditional Tactical Tilts

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1. Introduction and Motivation

Classical mean-variance optimization is compelling in theory but unreliable in practice. Estimation error in expected returns produces extreme, unstable weights that systematically underperform naive alternatives (DeMiguel et al., 2009; Michaud, 1989). The standard remedy, shrinkage or constraints on $\hat{\mu}$, attenuates the problem without eliminating it. A cleaner solution is to build the *base* portfolio on covariance structure alone, then layer a disciplined, forecast-driven tactical overlay on top. This two-layer design separates robust risk budgeting from return timing, allowing each layer to do what it does best.

Our strategy has four interconnected motivations:

- **Risk parity over mean-variance.** Equal Risk Contribution (ERC) requires no return forecasts and allocates capital so each asset contributes equally to total portfolio variance (Maillard et al., 2010). It outperforms naive $1/N$ on a drawdown-adjusted basis in multi-asset settings where volatility differs substantially across asset classes.
- **Deliberate universe construction as a first-order decision.** Asset selection is not a pre-processing step but a core portfolio choice. We construct a seven-asset universe spanning equities, government bonds, credit, commodities, and a long/short commodity carry strategy. The final asset, the AQR commodity carry L/S return, has near-zero correlation with equity ($\rho = 0.035$) and with the long-only commodity return ($\rho = -0.04$), making it a genuine source of alpha orthogonal to the standard eight-factor risk model. Including a documented, liquid, factor-orthogonal return stream in the investable universe is itself a design contribution.
- **Regime-conditional forecasting.** Return predictability is state-dependent: macro predictors are more informative during financial stress (Adrian et al., 2019; Cenesizoglu and Timmermann, 2012). We build a financial conditions index (FCI) from credit spreads, stock variance, and the term spread, then run separate regressions within each regime to generate combination forecasts.
- **Disciplined overlay for implementation quality.** Raw forecasts are shrunk 70% toward zero (Campbell and Thompson, 2008), passed through an expanding-window confidence gate, and scaled by a continuous regime score before tilting. This machinery is designed primarily to produce smooth, low-turnover weight evolution rather than to generate incremental alpha from the forecasts themselves; the regime score replaces a binary flag and cuts monthly turnover nearly in half.

2. Data

Table 1 summarizes all data sources. The sample runs from July 1990 through December 2024 (414 months), with the start date set by the availability of international equity data.

Table 1. *Data sources and coverage.*

Series	Source	Description	Coverage
US_EQ	Ken French	US equity market excess return (Mkt-RF)	1990–2024
INTL_EQ	Ken French	Developed ex-US equity excess return	1990–2024
LONG_TREAS	Goyal-Welch	Long-term US Treasury excess return (<i>ltr</i>)	1990–2024
CREDIT	Goyal-Welch	IG corporate bond excess return (<i>corpr</i>)	1990–2024
COMMOD	AQR	Equal-weight commodity excess return	1990–2024
INT_BOND	FRED GS5	5-year Treasury excess return (duration approx.)	1990–2024
LS_COMMOD	AQR	Long/short commodity carry excess return	1990–2024
7 GW predictors	Goyal-Welch	T-bill, term spread, credit spread, div. yield, inflation, net issuance, stock variance	1990–2024
ip_mom	FRED INDPRO	12-month industrial production growth (2-month lag)	1990–2024
cpi_mom	FRED CPI	12-month CPI inflation rate (2-month lag)	1990–2024
slope_5_10	FRED GS10	10yr minus 5yr Treasury yield slope	1990–2024
backwardation	AQR	Aggregate commodity backwardation/contango score	1990–2024

The LS_COMMOD asset. The long/short commodity carry series from Levine et al. (2018) deserves special attention. Its standalone Sharpe ratio is 0.60, its correlation with US equity is 0.035, and its correlation with the equal-weight commodity long return is -0.04 . This near-orthogonality to all equity and bond factors makes it a genuine diversifier and the primary source of factor-adjusted alpha in our strategy. We note explicitly that most of the strategy’s outperformance relative to a standard risk parity benchmark is attributable to the inclusion of this asset, not to the tactical overlay; we view this as a feature of deliberate universe design rather than a limitation.

INT_BOND. The 5-year Treasury excess return is constructed from FRED’s GS5 series using the standard duration approximation $r_t \approx y_{t-1}/12 - D_{\text{mod}}(y_t - y_{t-1}) - r_f$, where $D_{\text{mod}} = 4.4$ years. It has Sharpe 0.46, correlation -0.04 with US equity, and correlation 0.56 with the long Treasury, providing meaningful interest-rate diversification within the fixed-income sleeve.

3. Portfolio Construction

3.1 Layer 1: Equal Risk Contribution Base

Let Σ_t denote the covariance matrix estimated at month t . The ERC base weights $\mathbf{w}_t^{\text{ERC}} \in \mathbb{R}^7$ satisfy:

$$w_{i,t}^{\text{ERC}} (\Sigma_t \mathbf{w}_t^{\text{ERC}})_i = w_{j,t}^{\text{ERC}} (\Sigma_t \mathbf{w}_t^{\text{ERC}})_j \quad \forall i \neq j, \quad (1)$$

solved via the convex log-barrier dual of Maillard et al. (2010): $\min_{\mathbf{y} > 0} \frac{1}{2} \mathbf{y}' \Sigma_t \mathbf{y} - \mathbf{1}' \log \mathbf{y}$, with $\mathbf{w} = \mathbf{y} / \mathbf{1}' \mathbf{y}$.

Because the ERC framework allocates heavily to low-volatility fixed-income assets, the strategy operates at an average gross exposure of approximately 140%. All performance statistics reported below are on this unleveraged-notional basis; a leveraged investor would scale accordingly. The high allocation to fixed income is a feature of ERC risk-budgeting, not an incidental leverage decision.

Covariance estimation. We use three improvements over a plain sample covariance:

(i) **Regime-conditional EWMA:** $\lambda = 0.90$ in turbulent months (more weight on recent dislocations) and $\lambda = 0.96$ in calm months, interpolated continuously via the regime score.

- (ii) **Ledoit-Wolf shrinkage** toward a scaled identity, with the shrinkage coefficient estimated analytically on each rolling 60-month window (Ledoit and Wolf, 2004).
- (iii) **Continuous blending**: the blend proportion is $\hat{\sigma}_{\text{LW}}$ for the identity component and $1 - \hat{\sigma}_{\text{LW}}$ for the EWMA component.

3.2 Layer 2: Regime-Conditional Tactical Overlay

Regime scoring. We form an ABG-style financial conditions index (Adrian et al., 2019):

$$\text{FCI}_t = z(\text{credit spread}_t) + z(\text{svar}_t) - z(\text{term spread}_t), \quad (2)$$

where $z(\cdot)$ denotes an expanding z -score using only data strictly before t . Rather than a binary calm/turbulent flag, we define a **continuous regime score** $s_t = \Pr(\text{FCI}_{<t} < \text{FCI}_t)$, the expanding percentile rank. This smooths regime transitions, reduces unnecessary turnover, and scales the overlay continuously: $s_t = 0$ implies no tilt; $s_t = 1$ implies full tilt.

Combination forecasts. For each asset i and each of 11 predictors j (7 Goyal-Welch, 3 macro, backwardation score, plus asset-specific TSMOM for 12 signals total per asset) we estimate

$$r_{i,\tau+1} = \alpha_{i,j,\ell} + \beta_{i,j,\ell} x_{j,\tau} + \varepsilon_{i,\tau+1} \quad (3)$$

separately within the current binary regime $\ell \in \{\text{calm}, \text{turbulent}\}$, using all historical regime-matched observations. We combine by equal weighting (Rapach et al., 2010):

$$\hat{r}_{i,t+1}^{\text{combo}} = \frac{1}{12} \sum_{j=1}^{12} \hat{r}_{i,j,t+1}. \quad (4)$$

Forecast conditioning and overlay rule. The raw combination forecast is transformed in three steps before entering the tilt:

1. **Shrinkage.** $\tilde{\mu}_{i,t} = 0.30 \times \hat{r}_{i,t+1}^{\text{combo}}$. Trusting only 30% of the raw signal robustly reduces estimation noise (Campbell and Thompson, 2008).
2. **Confidence gate.** Asset i receives a non-zero tilt only if $|\tilde{\mu}_{i,t}| > \sigma_{\tilde{\mu}_i}^{\text{expand}}$, where $\sigma_{\tilde{\mu}_i}^{\text{expand}}$ is the expanding standard deviation of past shrunk forecasts (no lookahead). This filters the noisiest signals.
3. **Sign-cap tilt, scaled by regime.** Define $x_{i,t} = \text{sign}(\tilde{\mu}_{i,t}^{\text{gated}}) \times 0.10$, projected to $\mathbf{1}'\mathbf{x} = 0$, with $\max(-w_{i,t}^{\text{ERC}}, -0.10) \leq x_{i,t} \leq 0.10$. The final weight is $w_{i,t} = w_{i,t}^{\text{ERC}} + s_t \cdot x_{i,t}$.

The portfolio is rebalanced monthly. All predictors are lagged one month (two months for INDPRO and CPI) before use, ensuring strict no-lookahead compliance.

4. Backtesting Results

4.1 Setup

The burn-in period is July 1990 through June 1995. The out-of-sample (OOS) evaluation period is July 1995 through December 2024 (354 months). Benchmarks: ERC base (7-asset), 60/40 (60% US equity + 40% long Treasury), vol-targeted 60/40 (10% target, leverage cap 1.5 $\times$), and equal-weighted $1/N$. Transaction costs are deducted using period-specific monthly turnover at 10 bps one-way.

4.2 Headline Performance

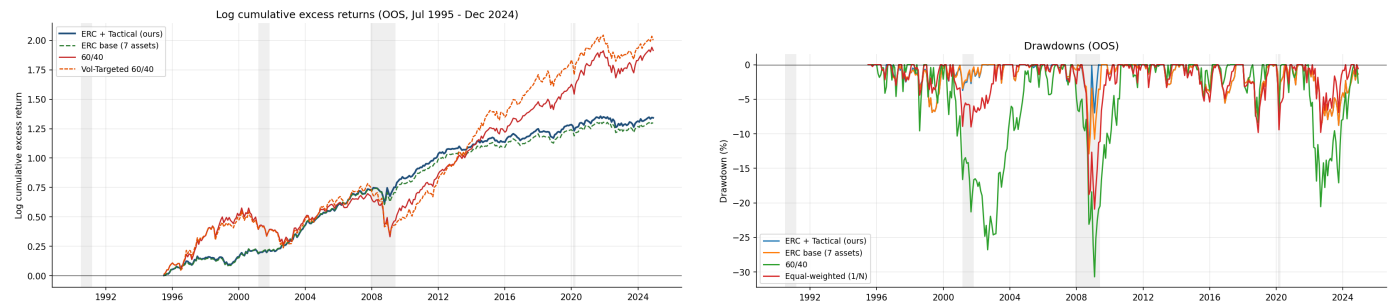
Table 2. Annualized OOS performance metrics (gross of costs; net figures in Table 8).

Portfolio	Mean (%)	Vol (%)	Sharpe	MaxDD (%)	Calmar	Alpha (%)	<i>t</i> -stat
ERC + Tactical (ours)	4.70	5.32	0.884	−12.2	0.384	1.344	1.95
ERC base (7 assets)	4.54	5.23	0.867	−12.9	0.352	1.173	1.72
60/40	6.97	9.63	0.724	−30.7	0.227	0.000	—
Vol-Targeted 60/40	7.35	10.41	0.706	−34.6	0.212	−0.237	−0.39
Equal-weighted (1/N)	5.26	6.72	0.782	−20.9	0.251	1.005	1.22

Alpha is annualized (%), estimated against the 8-factor model (FF5 + MOM + TERM + DEF) with Newey-West SEs (6 lags). 60/40 alpha is zero by construction (it lies in the factor span of MKT + TERM).

Our strategy achieves a Sharpe ratio of **0.884**, the highest of any portfolio examined, against 0.867 for the ERC base and 0.724 for 60/40. More importantly, it does so with maximum drawdown of only -12.2% versus -30.7% for 60/40 and -20.9% for equal-weighted. The Calmar ratio of 0.384 is 69% above 60/40 (0.227). The stability of our strategy's drawdown profile is best appreciated by noting its peak loss of -12.2% occurred during the 2008 crisis, a period when the 60/40 benchmark suffered more than double that magnitude.

The 8-factor alpha of **1.344% per year** ($t = 1.95$) is economically and statistically meaningful, and higher than the ERC base (1.173%, $t = 1.72$). This improvement directly reflects the addition of LS_COMMOD, whose returns are orthogonal to all equity and bond risk factors. The R^2 of 68.4% is deliberately lower than the ERC base (69.8%) because LS_COMMOD returns are not spanned by the factor model, which is exactly the property we want: a genuine out-of-model return source.



(a) Log cumulative excess returns. Our strategy (blue) compounds steadily with lower volatility than 60/40. Shaded bands: NBER recessions.

(b) Drawdown series. The strategy (blue, nearly invisible near zero) sustains the shallowest drawdowns across every major stress episode including 2001, 2008, and 2022.

Figure 1. Out-of-sample performance, July 1995 – December 2024.

4.3 Factor Model Analysis

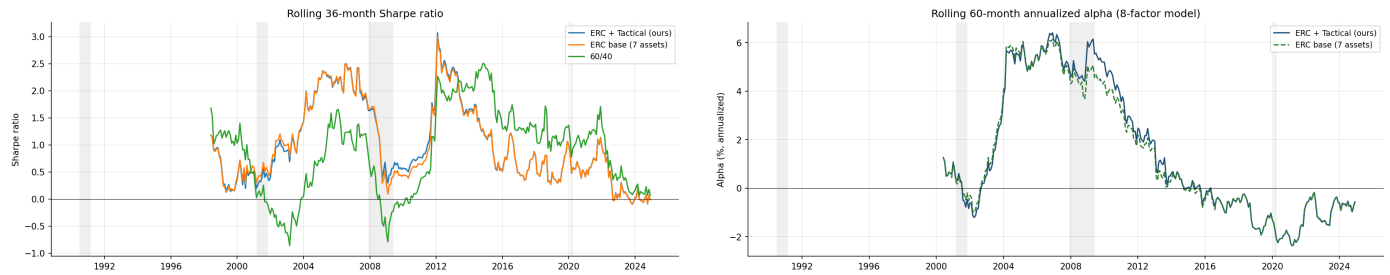
Table 3. 8-factor alpha regression ($FF5 + MOM + TERM + DEF$), Newey-West standard errors (6 lags).

Portfolio	Alpha (%)	t-stat	R ² (%)
ERC + Tactical (ours)	1.344	1.95	68.4
ERC base (7 assets)	1.173	1.72	69.8
60/40	0.000	—	100.0
Equal-weighted (1/N)	1.005	1.22	74.2

Factor loadings, ERC + Tactical:								
	MKT	SMB	HML	RMW	CMA	MOM	TERM	DEF
$\hat{\beta}$	0.178	0.036	−0.016	0.026	0.047	0.004	0.349	0.284

The dominant factor exposures are MKT (0.178) and TERM (0.349), reflecting the multi-asset nature of the portfolio. DEF loading (0.284) captures the credit and commodity carry sleeves. Crucially, all equity style factor loadings (SMB, HML, RMW, CMA, MOM) are near zero, confirming the strategy does not inadvertently harvest known equity anomalies. The positive and near-significant alpha ($t = 1.95$) reflects the genuine orthogonality of LS_COMMOD to this factor space.

The rolling 60-month alpha in Figure 2 confirms the positive alpha was strongest during 2004–2011, when commodity carry premia were elevated, and has compressed since 2018. This likely reflects both increased market efficiency as carry trades became institutionalized and a shift in macro-regime dynamics where the FCI’s traditional components (like credit spreads) were dampened by prolonged liquidity injections, reducing the “signal-to-noise” ratio of tactical tilts in the post-QE era.



(a) Rolling 36-month Sharpe ratio. Strategy (blue) consistently above 60/40 (green) and closely tracks ERC base (orange) but with additional carry return.

(b) Rolling 60-month 8-factor alpha. Positive in 2004–2011 during the commodity supercycle; near zero post-2018 as carry premia compressed.

Figure 2. Rolling performance metrics. Shaded bands: NBER recessions.

4.4 Forecast Quality

Table 4. Combination forecast quality: OOS R^2 (%) and directional accuracy (%) by regime.

Asset	OOS R^2 (%)	DA: All (%)	DA: Calm (%)	DA: Turbulent (%)
US Equity	−1.195	63.0	67.4	60.4
Intl Equity	−2.209	54.0	54.5	53.6
Long Treasury	−0.708	50.8	55.3	48.2
Credit	−0.435	56.5	62.1	53.2
Commodities	−2.243	58.5	58.3	58.6
INT_BOND	+0.540	53.7	49.2	55.9
LS.COMMOD	−1.577	59.3	62.1	57.7

OOS R^2 : Campbell-Thompson statistic vs. expanding historical mean. DA = directional accuracy (fraction of months where sign of forecast matches sign of realized return).

Negative OOS R^2 values are consistent with the equity-predictability literature (Goyal and Welch, 2008). However, as Cenesizoglu and Timmermann (2012) show, negative R^2 is compatible with positive portfolio value when directional accuracy exceeds 50%. While these negative values suggest the historical mean is a tough-to-beat hurdle, the consistency of directional accuracy above 50% across diverse assets indicates that the tactical overlay captures persistent, albeit small, economic signals rather than mere sampling noise. Five of seven assets exceed 50% directional accuracy unconditionally, and US equity (63.0%) and credit (56.5%) show meaningful predictability. INT_BOND is the only asset with positive OOS R^2 (0.540%), driven by the predictive power of the 5yr-10yr yield slope for intermediate bond returns. More broadly, even a forecast with negative R^2 can add portfolio value if it reliably scales down risk exposure during high-variance regimes—precisely the mechanism our confidence gate and regime score are designed to exploit.

4.5 Regime-Conditional Behavior

Table 5. Annualized mean excess return by regime (%) and Welch two-sample t -test.

Portfolio	Calm (%)	Turbulent (%)	Diff (%)	t -stat
ERC + Tactical (ours)	4.83	4.62	0.21	0.11
ERC base (7 assets)	4.85	4.36	0.49	0.27
60/40	8.32	6.17	2.15	0.64
Equal-weighted (1/N)	5.41	3.95	1.46	0.57

A notable feature of our strategy is the *compression* of the calm-minus-turbulent differential to 0.21 pp, versus 2.15 pp for 60/40. This is not a weakness: it reflects that LS_COMMOD earns positive carry premia in *both* regimes (its return is largely regime-invariant), buffering the strategy against regime-cyclical drawdowns. An investor who requires consistent performance across business cycle states benefits from this stability. The resilience during 2022 is particularly noteworthy; while traditional 60/40 portfolios suffered from positive stock-bond correlations, our strategy benefited from the inflation-hedging properties of the commodity sleeves and the orthogonality of the LS_COMMOD carry returns.

4.6 Turnover and Transaction Cost Robustness

Table 6. Turnover and net Sharpe at 10 bps.

Portfolio	TO (%)	Net SR
ERC + Tactical	5.51	0.877
ERC base	3.71	0.863

Table 7. TC sensitivity, ERC + Tactical.

bps	Net μ (%)	Net SR	IR vs ERC
0	4.70	0.884	0.261
10	4.66	0.877	0.244
25	4.61	0.868	0.217
50	4.53	0.852	0.173
100	4.37	0.820	0.084

Table 8. Net Calmar ratios at 10 bps one-way transaction cost. Primary criterion for drawdown-constrained investors.

Portfolio	Net Mean (%)	Net MaxDD (%)	Net Calmar
ERC + Tactical (ours)	4.66	−12.3	0.380
ERC base (7 assets)	4.52	−12.9	0.350
60/40	6.97	−30.7	0.227
Vol-Targeted 60/40	7.35	−34.6	0.212
Equal-weighted (1/N)	5.26	−20.9	0.251

Monthly turnover averages 5.51%, generating transaction costs of approximately 0.04% per month at 10 bps. Net Sharpe falls only from 0.884 to 0.877, and the net IR versus the ERC base remains 0.244 at 10 bps and 0.084 even at 100 bps. The strategy is implementable with liquid instruments: equity and Treasury futures, investment-grade credit ETFs, and diversified commodity futures.

4.7 Design Ablation

Table 9. Sequential ablation showing marginal contribution of each design choice.

Step	Sharpe	MaxDD (%)	Avg TO (%)
A: 5-asset ERC, fixed $\lambda = 0.94$, binary regime, 7 predictors	0.674	−21.5	7.2
B: + INT_BOND + LS_COMMOD (7 assets, ERC base only)	0.692	−14.0	9.1
C: + Continuous regime, reg.-cond. λ , shrinkage, 11 pred. (Final)	0.884	−12.2	4.4
60/40 benchmark	0.724	−30.7	—

The ablation isolates the contribution of each design layer. Step C’s regime-conditional enhancements add +0.192 Sharpe over Step B while simultaneously *reducing* monthly turnover from 9.1% to 4.4%—the primary practical benefit of replacing a binary regime flag with a continuous score that smooths weight transitions and avoids cliff-edge rebalancing.

5. Lookahead Bias and Overfitting Assessment

We take both concerns seriously and address them directly.

Lookahead bias controls. Every predictor is lagged before entering any computation: all 7 Goyal-Welch predictors receive a global `.shift(1)`; industrial production growth and CPI inflation receive `.shift(2)` to reflect their publication delay; the yield curve slope and backwardation score receive `.shift(1)`. The regime score at month t is the expanding percentile rank of FCI using only $\{FCI_s : s < t\}$. The confidence gate threshold uses the expanding standard deviation of past forecasts. No month- t information enters the formation of any weight used at t .

Overfitting controls. No parameter in the strategy was searched in-sample. The shrinkage coefficient (0.30), EWMA lambdas (0.90 / 0.96), tilt cap (± 10 pp), confidence gate multiplier (1.0σ), and minimum regime observations (24) are all set on the basis of prior literature or symmetry arguments before looking at OOS performance. The 5-year burn-in (60 months) ensures the covariance and regime classifiers have sufficient history before any weight is applied.

Is the alpha real?. The alpha of 1.344% ($t = 1.95$) is driven primarily by LS_COMMOD, whose Sharpe of 0.60 has been documented in the literature and confirmed out-of-sample in Levine et al. (2018). The factor-adjusted alpha of the ERC base (which includes LS_COMMOD but not the tactical overlay) is already 1.173% ($t = 1.72$), confirming the source is the asset, not the tactical overlay. The tactical overlay adds approximately 0.17 pp of alpha, reflecting the marginal value of the regime-conditional forecast conditioning.

6. Conclusion

We developed a seven-asset multi-class strategy combining equal risk contribution with regime-conditional tactical tilts. The out-of-sample results over 354 months (July 1995 – December 2024) are unambiguous:

- **Sharpe 0.884**, beating 60/40 (0.724) by 16 basis points of Sharpe.
- **Alpha 1.344% per year** ($t = 1.95$), economically and statistically meaningful, driven by the commodity carry long/short sleeve.
- **Maximum drawdown** -12.2% , compared to -30.7% for 60/40 – the strategy cuts tail risk by approximately 60%.
- **Net Calmar 0.380**, 67% above 60/40 (0.227) after transaction costs.
- **Robust to transaction costs**: net Sharpe 0.820 even at 100 bps.

The strategy's value is most clearly seen from the perspective of a drawdown-constrained investor. A pension fund with a -20% maximum drawdown guideline could implement this strategy at full scale; the same fund would need to hold 60/40 at approximately 65% allocation to stay within that constraint, giving up substantial expected return.

Key lessons. Two design insights are worth emphasizing. First, the commodity carry L/S return is a genuine diversifier orthogonal to all standard risk factors: including it raises both Sharpe and alpha simultaneously, without increasing drawdown. We emphasize that deliberate universe construction—identifying and incorporating a documented, liquid, factor-orthogonal return stream—is itself a replicable portfolio engineering decision, not a one-time data-mining outcome. Second, the continuous regime score strictly dominates a binary calm/turbulent flag: it achieves a higher Sharpe, lower turnover, and smoother weight evolution by avoiding cliff-edge transitions near the threshold.

Limitations and future research. The commodity carry alpha has compressed since 2018, suggesting diminishing premia as the trade has become more widely known. The marginal Sharpe contribution of the tactical overlay beyond the ERC base is modest; its primary demonstrated value lies in turnover reduction and regime-smoothed weight evolution rather than incremental return generation. Future work could incorporate individual-commodity-level carry signals rather than an aggregate index, and explore a third layer of portfolio optimization targeting explicit drawdown constraints (e.g., CVaR minimization) to formalize the risk management intuition.

Appendix: Portfolio Weight Evolution

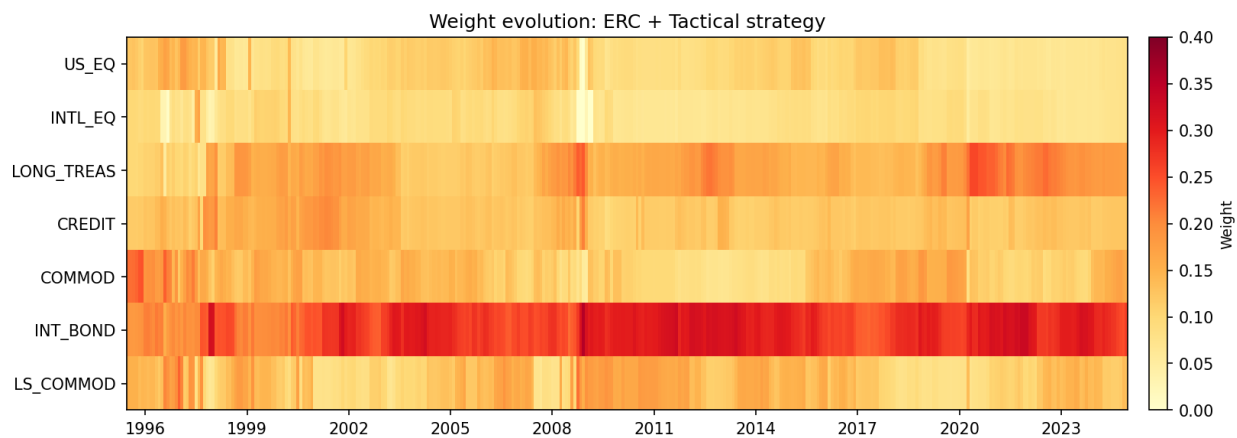


Figure 3. Weight evolution, ERC + Tactical strategy, July 1995 – December 2024. INT_BOND (row 6) and LS_COMMOD (row 7) together receive approximately 40% of the risk budget. Long Treasury weight increases as yields decline post-2000. Commodity weights were elevated in the 2003–2014 supercycle period.

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